RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering (Cyber Security)

PROJECT DOMAIN: WEB PROGRAMMING

PROJECT NAME: CAMPUS LOST AND FOUND PORTAL

A Web-Based Platform for Reporting and Recovering Lost Belongings

ABSTRACT

- The Campus Lost & Found Portal is a web-based application developed to simplify the process of reporting, searching for, and recovering misplaced belongings within a college campus. Traditional methods of reporting lost items often rely on notice boards and word of mouth, making information difficult to access.

- The proposed system provides a centralized platform where users can report lost or found items, upload photographs, and view details such as item descriptions, categories, locations, dates, and recovery status. The application improves accessibility, organizes item-related information, and helps students identify and recover their belongings more efficiently.

PROBLEM / OBJECTIVE

Students often face difficulties recovering misplaced belongings because lost-and-found information is scattered across different communication channels. Manual reporting makes searching, tracking, and updating item information inefficient.

- Develop a centralized platform for lost-and-found reports
- Enable users to submit detailed item descriptions and photographs.
- Simplify browsing and identification of reported belongings.
- Improve tracking of item status and recovery.
- Provide a user-friendly and accessible interface.

TOOLS, TECHNOLOGIES AND IMPLEMENTATION

Frontend

- React.js, Vite, JavaScript, HTML5, CSS3, React Router, Lucide React

Backend

- Python, Flask REST API, Flask-SQLAlchemy, Flask-Migrate, Flask-CORS, Werkzeug

Database

- MySQL, SQLAlchemy ORM, Environment variables for database credentials

Working

1. Users access the portal and browse existing listings.
2. Users submit details when reporting a lost or found item.
3. The backend processes the submitted information.
4. Item details and images are made available through the portal.
5. Users view relevant listings to help identify and recover belongings.

SOURCE CODE - Frontend

```jsx
function ScrollToTop() {
  const { pathname } = useLocation();
  useEffect(() => {
    window.scrollTo(0, 0);
  }, [pathname]);
  return null;
}

export default function App() {
  return (
    <AuthProvider>
      <Router>
        <ScrollToTop />
        <div style={{ display: 'flex', flexDirection: 'column', minHeight: '100vh' }}>
          <Navbar />
          <main style={{ flexGrow: 1 }}>
            <Routes>
              <Route path="/" element={<LandingPage />} />
              <Route path="/browse" element={<BrowseItemsPage />} />
              <Route path="/report" element={<ReportItemPage />} />
              <Route path="/items/:id" element={<ItemDetailPage />} />
              <Route path="/matches" element={<SmartMatchPage />} />
              <Route path="/dashboard" element={<DashboardPage />} />
              <Route path="/notifications" element={<NotificationsPage />} />
              <Route path="/admin" element={<AdminPage />} />
              <Route path="/login" element={<LoginPage />} />
              <Route path="/register" element={<RegisterPage />} />
              <Route path="*" element={<LandingPage />} />
            </Routes>
          </main>
          <Footer />
        </div>
      </Router>
    </AuthProvider>
  );
}
```

```jsx
const itemImages = {
  'Black Hydro Flask Water Bottle': '/images/hydroflask.jpg',
  'Matte Black Insulated Flask': '/images/flask.jpg',

  'Blue Campus Student ID Card': '/images/bluidcard.jpg',
  'Student ID Card with Blue Lanyard': '/images/idcard.jpg',

  'Apple AirPods Pro (2nd Gen)': '/images/airpods.jpg',
  'TI-84 Plus Graphing Calculator': '/images/calculator.jpg',

  'Spiral Chemistry Lab Notebook': '/images/notebook.jpg',
  'Navy Blue Nike Backpack': '/images/navybag.jpg',
};

export default function ItemCard({ item }) {
  const formattedDate = item.incident_date
    ? new Date(item.incident_date).toLocaleDateString('en-US', {
        month: 'short',
        day: 'numeric',
        year: 'numeric'
      })
    : 'Unknown date';

  // Prefer an uploaded image; otherwise use the sample image
  const imagePath = item.image_path || itemImages[item.item_name];
```

```jsx
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

```jsx
<div
  className="card item-card"
  style={{
    display: 'flex',
    flexDirection: 'column',
    height: '100%'
  }}
>
  {/* Image Area */}
  <div
    style={{
      position: 'relative',
      height: '190px',
      background: imagePath
        ? `url("${imagePath}") center / cover no-repeat`
        : 'linear-gradient(135deg, #1e1b4b 0%, #312e81 100%)',
      display: 'flex',
      alignItems: 'center',
      justifyContent: 'center',
      borderBottom: '1px solid var(--border-light)'
    }}
  >
    {!imagePath && (
      <div
        style={{
          textAlign: 'center',
          color: 'rgba(255, 255, 255, 0.4)'
        }}
      >
        <Package size={44} strokeWidth={1.5} />
        <div
          style={{
            fontSize: '0.75rem',
            marginTop: '4px',
            textTransform: 'uppercase',
            letterSpacing: '0.05em'
          }}
        >
          {item.category_name}
```

SOURCE CODE - Database

```sql
CREATE DATABASE IF NOT EXISTS findit_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE findit_db;

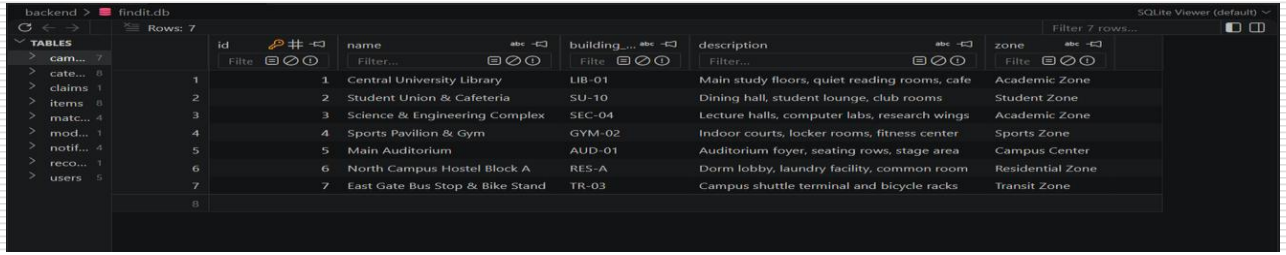
-- 1. USERS TABLE
CREATE TABLE IF NOT EXISTS users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(120) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) NOT NULL DEFAULT 'student', -- 'student', 'moderator', 'admin'
    student_id_number VARCHAR(50) NULL,          -- Private student identifier
    phone VARCHAR(30) NULL,                      -- Private phone
    account_status VARCHAR(20) NOT NULL DEFAULT 'active', -- 'active', 'suspended'
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_user_email (email)
) ENGINE=InnoDB;

-- 2. CATEGORIES TABLE
CREATE TABLE IF NOT EXISTS categories (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(80) NOT NULL UNIQUE,
    icon VARCHAR(40) NOT NULL DEFAULT 'Package',
    description VARCHAR(255) NULL
) ENGINE=InnoDB;

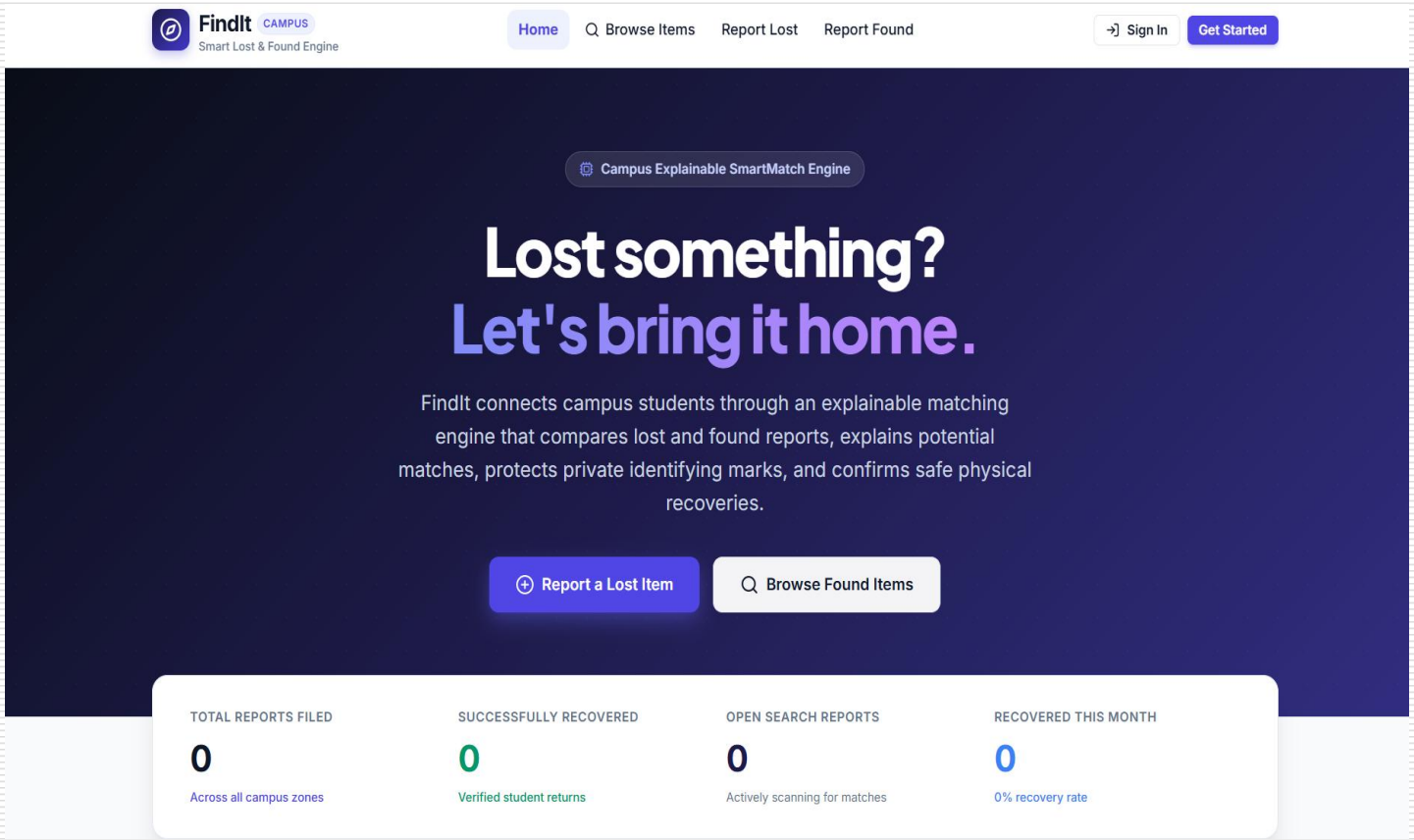
-- 3. CAMPUS LOCATIONS TABLE
CREATE TABLE IF NOT EXISTS campus_locations (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(120) NOT NULL UNIQUE,
    building_code VARCHAR(30) NULL,
    description VARCHAR(255) NULL,
    zone VARCHAR(50) NULL -- e.g. Academic Zone, Student Zone, Sports Zone
) ENGINE=InnoDB;
```

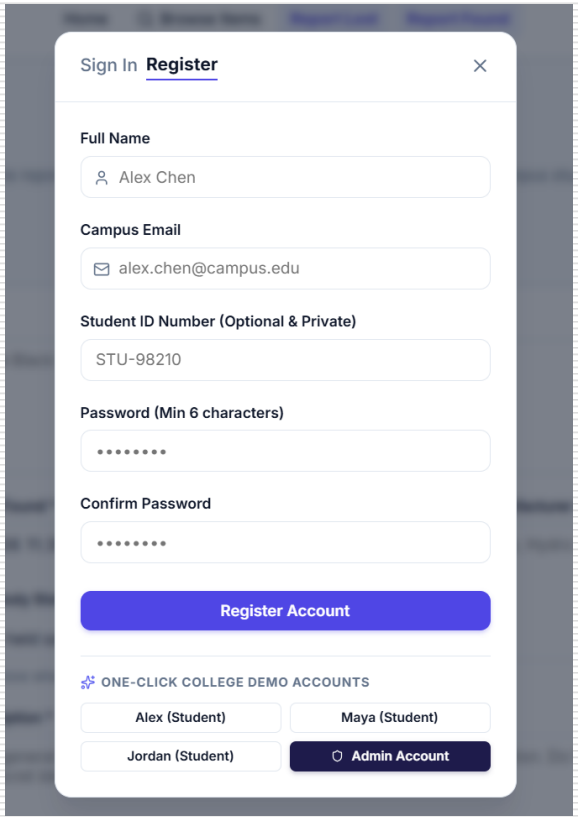
```sql
CREATE TABLE IF NOT EXISTS items (
    id INT AUTO_INCREMENT PRIMARY KEY,
    owner_user_id INT NOT NULL,
    report_type VARCHAR(10) NOT NULL,            -- 'lost' or 'found'
    item_name VARCHAR(150) NOT NULL,
    category_id INT NOT NULL,
    public_description TEXT NOT NULL,
    private_verification_details TEXT NULL,      -- Strictly private verification data
    color VARCHAR(50) NULL,
    brand VARCHAR(80) NULL,
    campus_location_id INT NOT NULL,
    incident_date DATETIME NOT NULL,
    image_path VARCHAR(255) NULL,
    custody_status VARCHAR(50) NOT NULL DEFAULT 'held_by_finder', -- 'held_by_finder', 'handed_to_campus_office', 'stored_in_depa
    status VARCHAR(30) NOT NULL DEFAULT 'open',  -- 'open', 'claim_pending', 'recovered', 'closed', 'withdrawn', 'flagged'
    contact_preference VARCHAR(30) NOT NULL DEFAULT 'in_app',
    is_flagged BOOLEAN NOT NULL DEFAULT FALSE,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_item_owner (owner_user_id),
    INDEX idx_item_type (report_type),
    INDEX idx_item_status (status),
    INDEX idx_item_category (category_id),
    INDEX idx_item_location (campus_location_id),
    CONSTRAINT fk_items_user FOREIGN KEY (owner_user_id) REFERENCES users(id) ON DELETE CASCADE,
    CONSTRAINT fk_items_category FOREIGN KEY (category_id) REFERENCES categories(id),
    CONSTRAINT fk_items_location FOREIGN KEY (campus_location_id) REFERENCES campus_locations(id)
) ENGINE=InnoDB;

-- 5. MATCHES TABLE (SmartMatch Explainable Results)
CREATE TABLE IF NOT EXISTS matches (
    id INT AUTO_INCREMENT PRIMARY KEY,
    lost_item_id INT NOT NULL,
    found_item_id INT NOT NULL,
    total_score FLOAT NOT NULL,
    factor_scores_json TEXT NOT NULL,            -- JSON string of individual criteria scores
    explanation_json TEXT NOT NULL,              -- JSON string of reasons, mismatches, and confidence
    status VARCHAR(30) NOT NULL DEFAULT 'suggested', -- 'suggested', 'reviewed', 'dismissed', 'claimed'
    algorithm_version VARCHAR(20) NOT NULL DEFAULT 'v1.0',
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
```

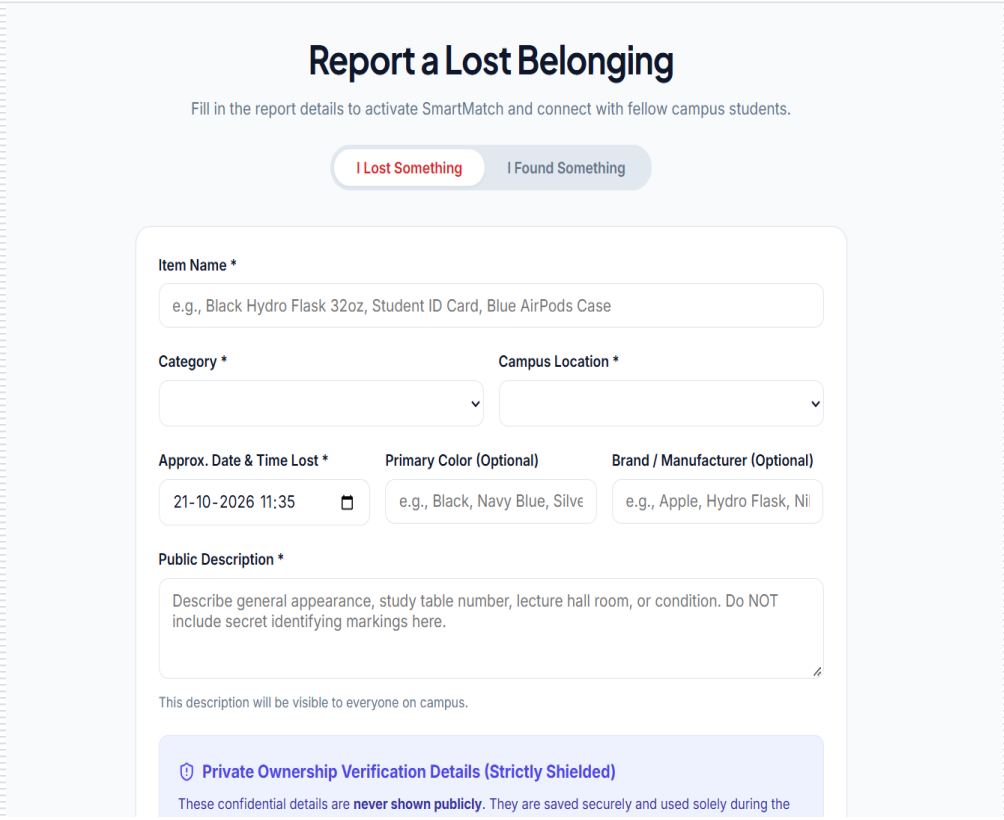


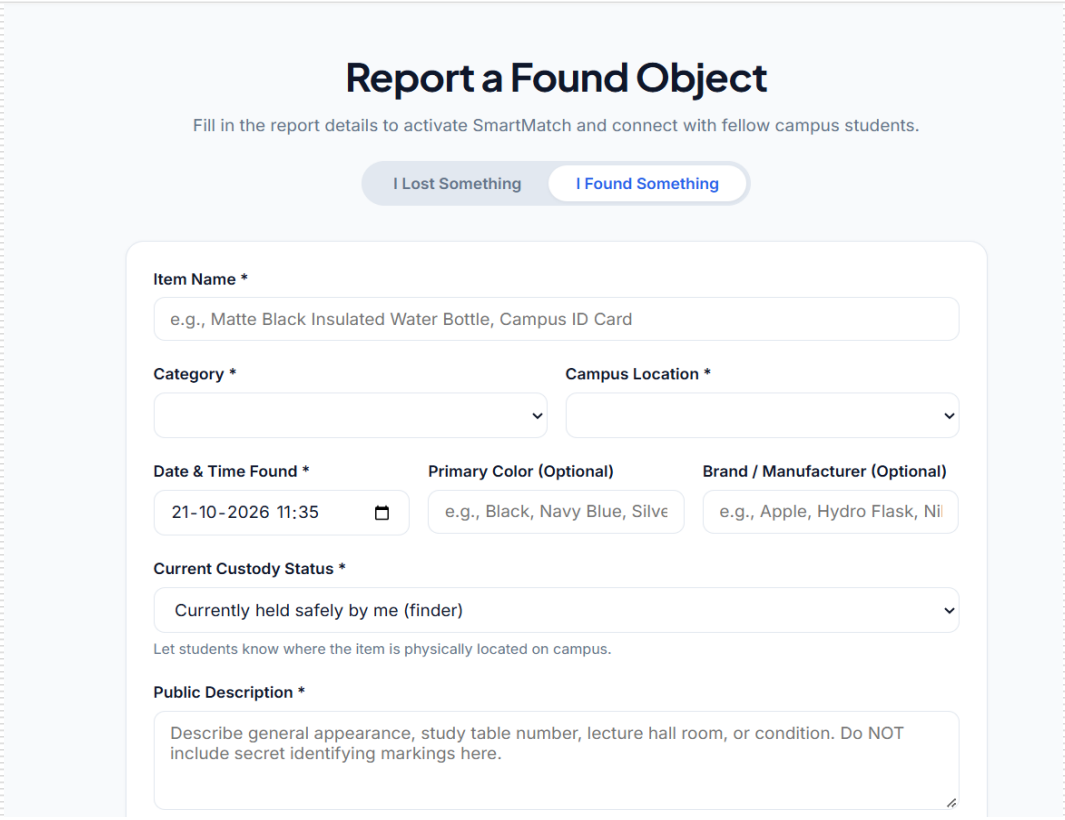
Website


FindIt CAMPUS
Smart Lost & Found Engine
Home
Browse Items
Report Lost
Report Found
Sign In
Get Started
Campus Explainable SmartMatch Engine
Lost something?
Let's bring it home.
FindIt connects campus students through an explainable matching engine that compares lost and found reports, explains potential matches, protects private identifying marks, and confirms safe physical recoveries.
Report a Lost Item
Browse Found Items
TOTAL REPORTS FILED
0
Across all campus zones
SUCCESSFULLY RECOVERED
0
Verified student returns
OPEN SEARCH REPORTS
0
Actively scanning for matches
RECOVERED THIS MONTH
0
0% recovery rate


Sign In
Register
Full Name
Alex Chen
Campus Email
alex.chen@campus.edu
Student ID Number (Optional & Private)
STU-98210
Password (Min 6 characters)
Confirm Password
Register Account
ONE-CLICK COLLEGE DEMO ACCOUNTS
Alex (Student)
Maya (Student)
Jordan (Student)
Admin Account

Website


Report a Lost Belonging
Fill in the report details to activate SmartMatch and connect with fellow campus students.
I Lost Something
I Found Something
Item Name *
e.g., Black Hydro Flask 32oz, Student ID Card, Blue AirPods Case
Category *
Campus Location *
Approx. Date & Time Lost *
21-10-2026 11:35
Primary Color (Optional)
e.g., Black, Navy Blue, Silve
Brand / Manufacturer (Optional)
e.g., Apple, Hydro Flask, Ni
Public Description *
Describe general appearance, study table number, lecture hall room, or condition. Do NOT include secret identifying markings here.
This description will be visible to everyone on campus.
Private Ownership Verification Details (Strictly Shielded)
These confidential details are never shown publicly. They are saved securely and used solely during the


Report a Found Object
Fill in the report details to activate SmartMatch and connect with fellow campus students.
I Lost Something
I Found Something
Item Name *
e.g., Matte Black Insulated Water Bottle, Campus ID Card
Category *
Campus Location *
Date & Time Found *
21-10-2026 11:35
Primary Color (Optional)
e.g., Black, Navy Blue, Silve
Brand / Manufacturer (Optional)
e.g., Apple, Hydro Flask, Ni
Current Custody Status *
Currently held safely by me (finder)
Let students know where the item is physically located on campus.
Public Description *
Describe general appearance, study table number, lecture hall room, or condition. Do NOT include secret identifying markings here.

CONCLUSION

The Campus Lost & Found Portal provides a convenient and organized solution for managing misplaced belongings within a college campus. By digitizing the reporting process and presenting item information in a centralized interface, the system reduces reliance on manual announcements and improves accessibility.